

NYC vs DC

BIKESHARE ANALYSIS



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DAB26



NYC

DC





AGENDA



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INTRODUCTION

Two iconic cities, two very different transit scales. This project compares the bikeshare networks of **New York City** and **Washington D.C.** to understand how urban pressure shapes the way people move. By putting both systems side by side, we identify the **biggest operational challenges** and focus our strategy to deliver **high impact solutions where they matter most.**



PROBLEM STATEMENT



During **2016 and 2017**, bikeshare usage showed a **clear difference in rider behavior** between **NYC** and **Washington DC**.

Subscribers accounted for **94.09%** of trips in **NYC** compared to **75.83%** in **DC**, indicating a higher proportion of casual riders in **DC**.

This **high volume of casual riders in DC** causes major **operational pressure**, leading to frequent **station overcrowding** and **higher maintenance costs**.





OBJECTIVES & HYPOTHESIS



OBJECTIVES



01

How does bikeshare **usage differ** between **weekdays and weekends** for subscribers and casual riders in **NYC** and **DC**?



02

When does bikeshare demand reach its **highest level** for subscribers and casual riders in **NYC** and **DC**?



03

Which **start and end stations** experience the **highest demand** in **NYC** and **DC**?



04

How does the **overall volume** of casual rider trips in **DC** compare to **NYC** across different **months or seasons**?



HYPOTHESIS



Casual riders in both **NYC** and **DC** have a **higher average trip duration** than subscribers.



ANALYSIS



High Weekday Subscriber Demand Drops on Weekends

Total Trips by Day Type and User Type

● Casual ● Subscriber

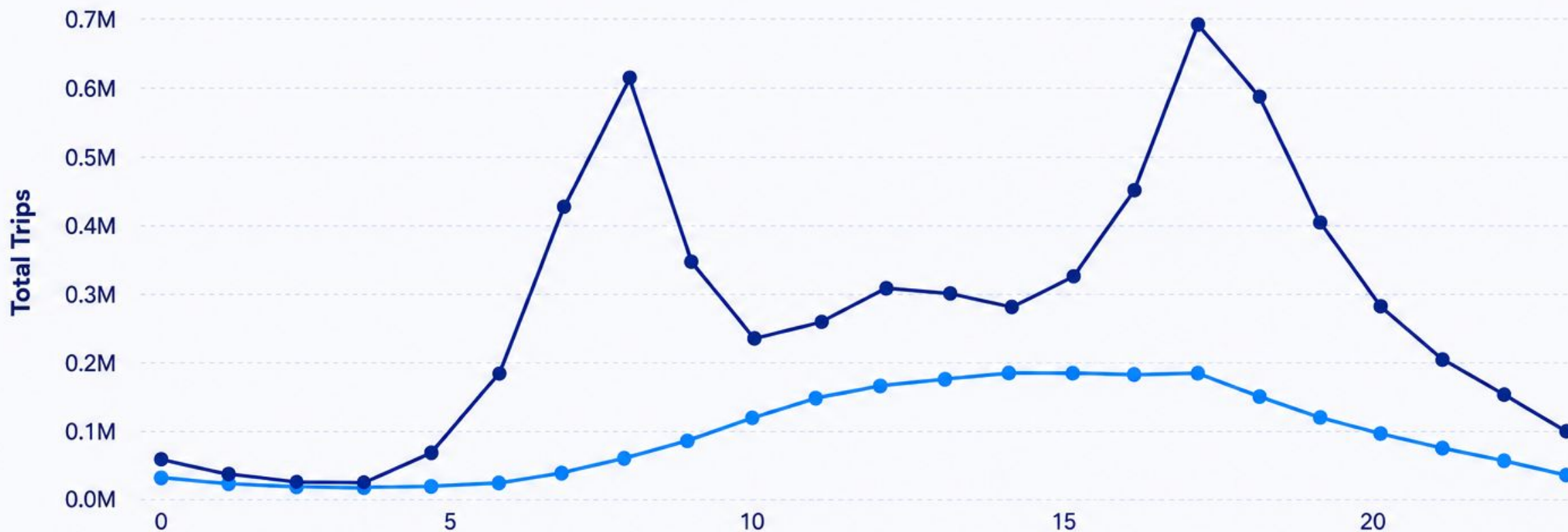




Bikeshare Demand Peaks During Morning and Evening Hours

Hourly Bikeshare Demand by User Type and City

user_type ● Casual ● Subscriber





ANALYSIS

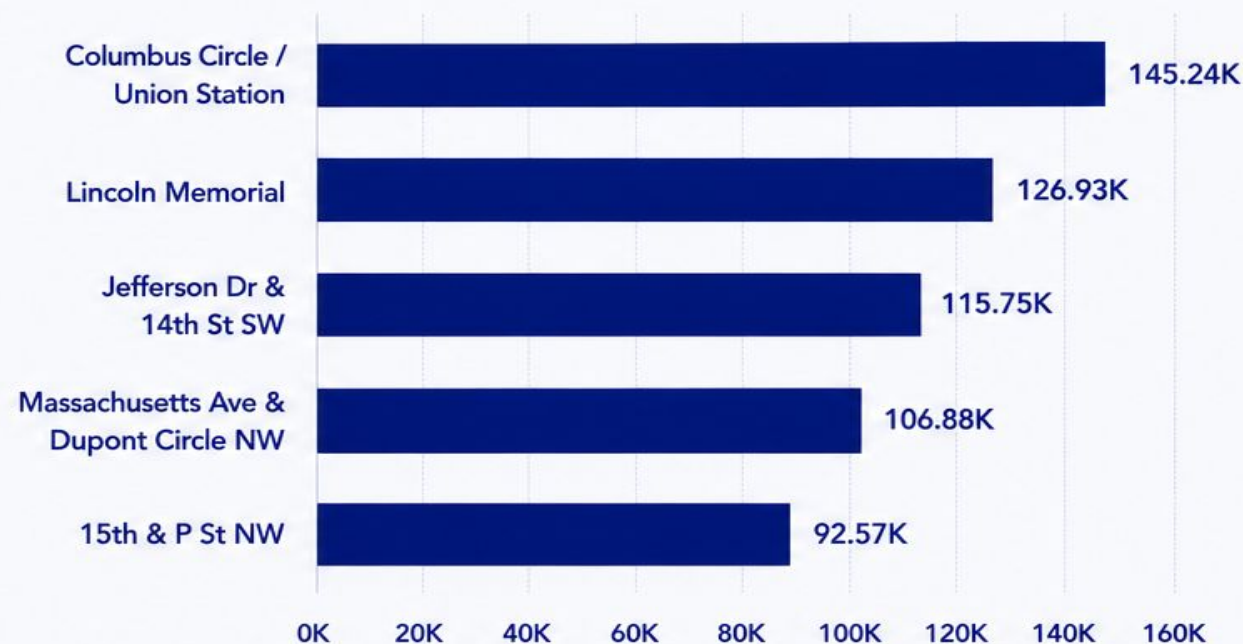


Most Popular Bike Stations in Washington D.C

Top 5 Start Stations



Top 5 End Stations





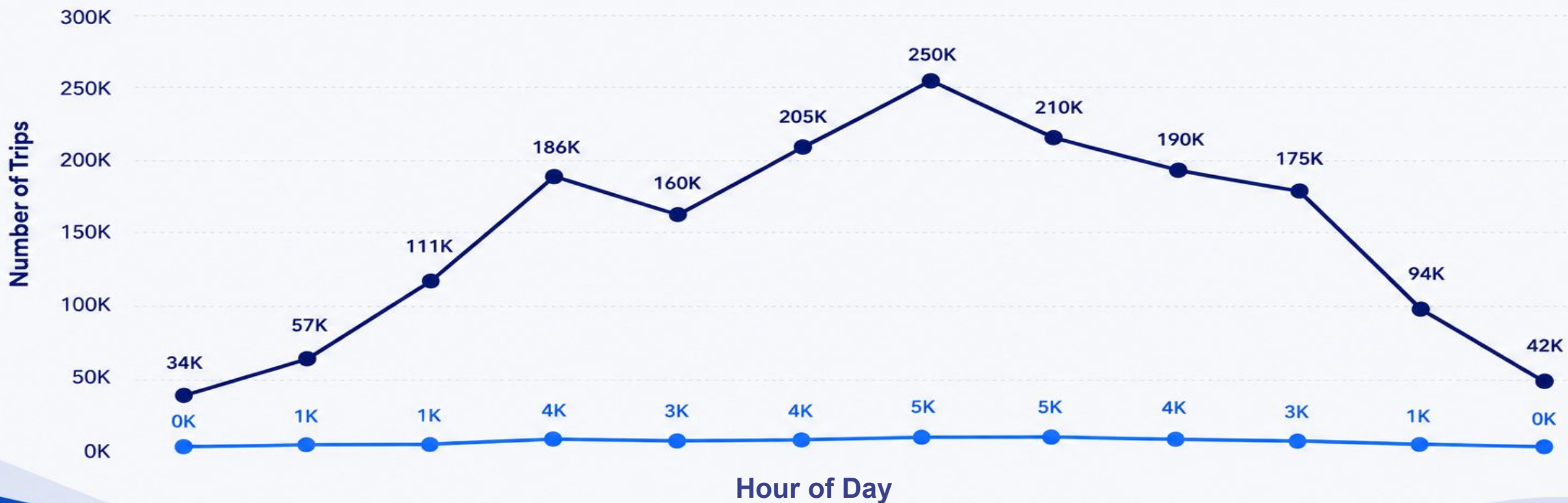
ANALYSIS



DC Casual Trips Peak in Summer While NYC Remains Flat

Casual Trip Volume by Month & City

—●— NewYorkCity-NVC —●— Washington-DC





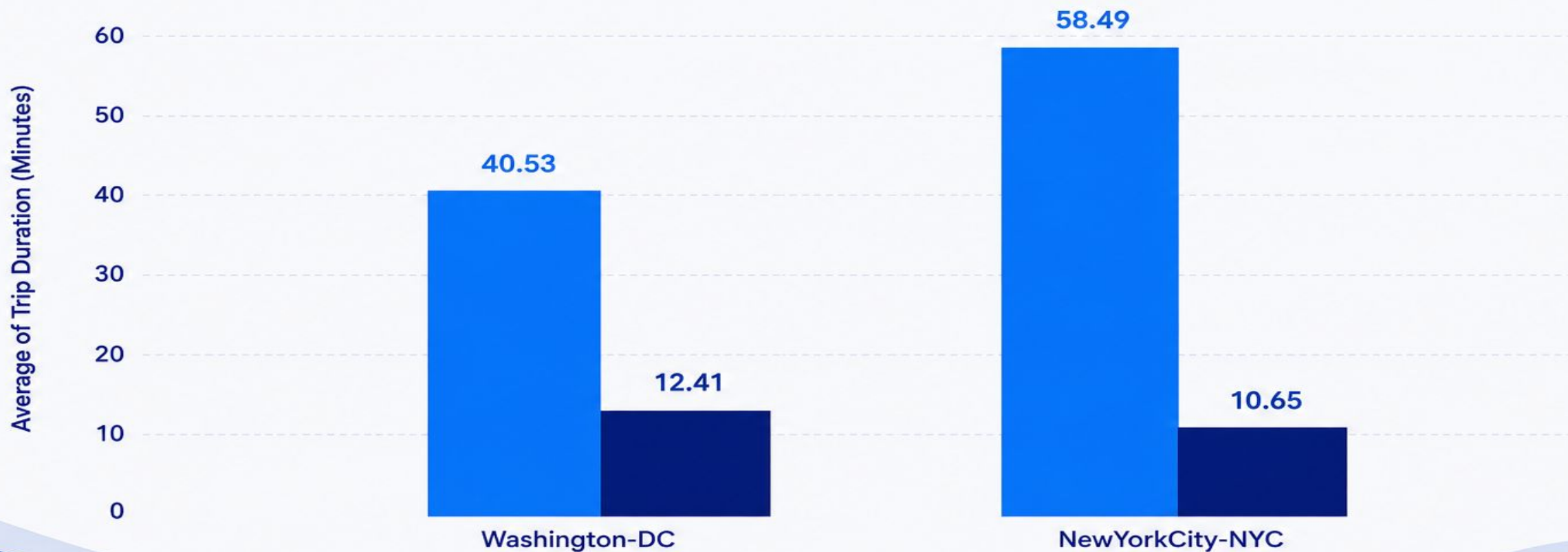
ANALYSIS



Casual Riders Take Longer Trips Than Subscribers

Average Trip Duration by User Type

● Casual ● Subscriber





RECOMMENDATIONS



01

Introduce a long ride pass.



02

Place station QR codes.



03

Launch weekend memberships.



04

Send targeted app alerts.



05

Optimize mid year rebalancing.



06

Offer user ride credits.





LIMITATIONS



01

Historical data may not **reflect current patterns.**



02

Demographic data like **gender** is **inconsistent.**



03

Financial records on **revenue and costs** unavailable.



04

Weather variations and **seasonal changes** not deeply covered.



05

External factors like **competing transport choices** excluded.